Atomic Physics: Worked Problems

Constants

$$R_\infty = 1.0974 \times 10^7\,\mathrm{m}^{-1}, \qquad m_e/m_p = 1/1836, \qquad a_0 = 5.29 \times 10^{-11}\,\mathrm{m},$$
$$\alpha \approx 1/137, \qquad e^2/4\pi\varepsilon_0 = \hbar^2/(m_e a_0), \qquad \mathrm{Ry} = 13.6\,\mathrm{eV}.$$

1 Spectral lines and isotope effects

Problem 1: H–D isotope shift

Find the wavelength difference between the Balmer-α ($n = 3 \to 2$) lines of hydrogen and deuterium.

Solution. With a finite nuclear mass M, the Rydberg constant becomes $R_M = R_\infty/(1 + m_e/M)$. For $n = 3 \to 2$,

$$\frac{1}{\lambda} = R_M\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5}{36}R_M \quad \Rightarrow \quad \lambda = \frac{36}{5R_M} = \frac{36}{5R_\infty}\left(1 + \frac{m_e}{M}\right).$$

Subtracting the two isotopes and using $m_D \approx 2m_H$:

$$\lambda_H - \lambda_D = \frac{36\,m_e}{5R_\infty}\left(\frac{1}{m_H} - \frac{1}{m_D}\right) \approx \frac{18}{5}\frac{m_e}{m_H}\frac{1}{R_\infty} = \frac{3.6}{1836 \times 1.0974 \times 10^7\,\mathrm{m}^{-1}} \approx 0.18\,\mathrm{nm}.$$

Problem 2: He^+ lines versus hydrogen lines

Show that certain He^+ lines nearly coincide with hydrogen lines, and estimate the wavelength ratio.

Solution. For a hydrogen-like ion, $1/\lambda = Z^2 R_M(1/m^2 - 1/n^2)$. With $Z = 2$ and the principal numbers doubled ($m = 2m'$, $n = 2n'$),

$$Z^2\left(\frac{1}{4m'^2} - \frac{1}{4n'^2}\right) = \frac{1}{m'^2} - \frac{1}{n'^2},$$

so each He^+ line sits close to a hydrogen line with the same m', n'. They differ only through the reduced-mass factor:

$$\frac{\lambda_H}{\lambda_{He}} = \frac{R_{He}}{R_H} = \frac{1 + m_e/m_H}{1 + m_e/m_{He}} \approx 1 + \frac{m_e}{m_H}\left(1 - \frac{1}{4}\right) = 1 + \frac{0.75}{1836} \approx 1.00041.$$

Problem 3: Resolving the fine structure

Estimate the resolving power needed to see the $n = 2$ fine structure of hydrogen in the Balmer-α line, and the number of grating lines required.

Solution. The $n = 2$ fine-structure splitting is of order $\alpha^2|E_2|/4$:

$$\Delta E \approx \frac{(1/137)^2 \times 3.4\,\text{eV}}{4} \approx 4.5 \times 10^{-5}\,\text{eV}.$$

The line energy is $\approx 1.9\,\text{eV}$, so

$$\frac{\lambda}{\Delta\lambda} = \frac{E}{\Delta E} \approx 4 \times 10^4.$$

Since the resolving power of a grating is mN (m the order, N the number of illuminated lines), $N \sim 10^5$ lines in first order is enough (to order of magnitude).

2 X-ray transitions

Problem 4: Moseley's law

The K_α frequency is $f = \frac{3}{4}cR_\infty(Z - \sigma)^2$ with screening constant $\sigma \approx 1$. Estimate the K_α energy of iron ($Z = 26$) and of lead ($Z = 82$), and comment on how $\sqrt{f}$ scales with Z.

Solution. The photon energy is

$$E_{K\alpha} = \tfrac{3}{4}\,(13.6\text{ eV})\,(Z-1)^2 \approx 10.2\,(Z-1)^2\text{ eV}.$$

For Fe: $10.2 \times 25^2 \approx 6.4\,\text{keV}$. For Pb: $10.2 \times 81^2 \approx 66.9\,\text{keV}$. For $Z \gg \sigma$, $f \propto Z^2$, so $\sqrt{f} \propto Z$, which gives a straight Moseley plot.

Problem 5: Size of the relativistic correction for lead

Estimate the L-shell energy of lead from the hydrogen-like formula and the scale of its fine-structure splitting relative to $E_{K\alpha}$.

Solution. For $Z = 82$, $n = 2$:

$$E_L \approx 13.6\,\frac{Z^2}{n^2}\text{ eV} = 13.6 \times \frac{82^2}{4} \approx 22.9\,\text{keV}.$$

Taking the hydrogen-like splitting scale $\Delta E \approx \alpha^2 E_L/4 \approx 0.3\,\text{keV}$ (crude, since screening is ignored),

$$\frac{\Delta E}{E_{K\alpha}} \approx \frac{0.3}{66.9} \approx 5 \times 10^{-3}.$$

Because $\Delta E \propto Z^4$ while $E_{K\alpha} \propto Z^2$, the relative correction grows as Z^2, which is why fine structure is conspicuous in heavy-atom X-ray lines.

3 Radiation

Problem 6: Radiative lifetime of a classical orbit

An electron in a circular orbit of radius r and angular frequency ω radiates classically. Estimate the lifetime $\tau = \hbar\omega/P$ for visible light of wavelength 650 nm with $r = a_0$.

Solution. A linear oscillation $x = r\cos\omega t$ radiates (time-averaged)

$$P_{\text{lin}} = \frac{e^2 r^2 \omega^4}{12\pi\varepsilon_0 c^3}.$$

A circular orbit is two perpendicular linear oscillations in quadrature, so the power doubles:

$$P = \frac{e^2 r^2 \omega^4}{6\pi\varepsilon_0 c^3} \quad \Rightarrow \quad \tau = \frac{\hbar\omega}{P} = \frac{6\pi\varepsilon_0 \hbar c^3}{e^2 r^2 \omega^3} \;\propto\; \frac{1}{r^2\omega^3}.$$

With $\omega = 2\pi c/\lambda = 2.9 \times 10^{15}\,\text{rad/s}$:

$$\tau = \frac{(1.67\times10^{-10})(1.05\times10^{-34})(2.69\times10^{25})}{(2.57\times10^{-38})(2.80\times10^{-21})(2.44\times10^{46})}\ \text{s} \approx 2.7\times10^{-7}\,\text{s}.$$

Problem 7: Population ratio and black-body radiation

A two-level system with degeneracies $g_1 = 1$, $g_2 = 3$ has $N_2 = 0.1N$ and $N_1 = 0.9N$, and the transition is at 600 nm. Find the temperature of the thermal radiation field that would produce this population, and the spectral energy density at the transition frequency.

Solution. Thermal equilibrium gives $N_2/N_1 = (g_2/g_1)e^{-\hbar\omega/kT}$, so

$$\frac{0.1}{0.9} = 3e^{-x} \;\Rightarrow\; e^x = 27, \qquad x = \frac{\hbar\omega}{kT} = \ln 27 \approx 3.30.$$

With $\omega = 2\pi c/\lambda = 3.14\times10^{15}\,\text{rad/s}$, $\hbar\omega = 3.31\times10^{-19}\,\text{J}$ (about 2.07 eV),

$$T = \frac{\hbar\omega}{kx} = \frac{3.31\times10^{-19}}{(1.38\times10^{-23})(3.30)} \approx 7.3\times10^{3}\,\text{K}.$$

Planck's law in angular frequency is

$$\rho(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3}\frac{1}{e^x - 1} = \frac{(1.05\times10^{-34})(3.10\times10^{46})}{\pi^2(2.69\times10^{25})}\cdot\frac{1}{26} \approx 4.7\times10^{-16}\,\text{J s m}^{-3}.$$

4 The Zeeman effect

Problem 8: Larmor frequency

Compute the Larmor frequency shift for $B = 3\times10^{-5}\,\text{T}$ (Earth's field) and $B = 1\,\text{T}$.

Solution. $\Omega_L = eB/2m_e = 8.79\times10^{10}\,\text{rad s}^{-1}\,\text{T}^{-1}\times B$.

B	Ω_L (rad/s)	$\Omega_L/2\pi$ (Hz)
$3\times10^{-5}\,\text{T}$	2.6×10^{6}	4.2×10^{5}
$1\,\text{T}$	8.8×10^{10}	1.4×10^{10}

Compared with an optical angular frequency of about $3\times10^{15}\,\text{rad/s}$, the fractional shifts are about 10^{-9} and 3×10^{-5} respectively.

Problem 9: Relative intensities in the normal Zeeman effect

Let the three eigen-oscillations of the electron each carry intensity I. Find (a) the total intensity radiated perpendicular to $\mathbf{B}$ and (b) the ratio of intensities observed along and perpendicular to $\mathbf{B}$ in each component.

Solution.

- Viewed along $\mathbf{B}$: only the circular motions in the plane perpendicular to $\mathbf{B}$ contribute, giving the two $\sigma^{\pm}$ lines, each with intensity I. The π (linear along $\mathbf{B}$) oscillation does not radiate along its own axis.
- Viewed perpendicular to $\mathbf{B}$: the π line is seen at full intensity I. Each circular motion is seen edge-on as a linear oscillation, so only a projected half of its intensity reaches the detector, giving $I/2$ in each σ line.

Therefore (a) the total perpendicular intensity is $I + I/2 + I/2 = 2I$, and (b) the perpendicular $\pi : \sigma^+ : \sigma^-$ pattern is $2 : 1 : 1$ (i.e. $I : \frac{1}{2}I : \frac{1}{2}I$), whereas along $\mathbf{B}$ it is $0 : 1 : 1$.

5 Bohr model and Rydberg atoms

Problem 10: Correspondence principle

Show that the quantum frequency for $n \to n-1$ at large n matches the classical orbital frequency, and that this implies $r = a_0 n^2$.

Solution. The total energy of a circular orbit is $E = -\frac{1}{2}m_e v^2 = -\dfrac{e^2}{8\pi\varepsilon_0 r}$. The classical orbital frequency satisfies

$$\omega_{cl}^2 = \frac{e^2}{4\pi\varepsilon_0\, m_e r^3} = \frac{\hbar^2}{m_e^2 a_0 r^3}.$$

Quantum mechanically, with $E_n = -\mathrm{Ry}/n^2$,

$$\omega_q = \frac{E_n - E_{n-1}}{\hbar} \approx \frac{2\,\mathrm{Ry}}{\hbar n^3} = \frac{\hbar}{m_e a_0^2 n^3},$$

using $\mathrm{Ry} = \hbar^2/(2m_e a_0^2)$. Setting $\omega_{cl} = \omega_q$:

$$\frac{\hbar^2}{m_e^2 a_0 r^3} = \frac{\hbar^2}{m_e^2 a_0^4 n^6} \;\Rightarrow\; r^3 = a_0^3 n^6 \;\Rightarrow\; r = a_0 n^2.$$

Problem 11: Rydberg atom, $n = 50$

Find the level spacing and the orbital radius for $n = 50$.

Solution. From $E_n = -\mathrm{Ry}/n^2$,

$$\Delta E \approx \left|\frac{\mathrm{d}E}{\mathrm{d}n}\right| = \frac{2\,\mathrm{Ry}}{n^3} = \frac{2(13.6)}{50^3}\ \mathrm{eV} \approx 2.2 \times 10^{-4}\,\mathrm{eV},$$

and

$$r = a_0 n^2 = 2500\, a_0 \approx 132\,\mathrm{nm}.$$

Rydberg atoms are therefore enormous (about 10^3 times larger in area scaling n^4 than the ground state) and have tightly packed levels.

6 Angular-momentum eigenfunctions

Problem 12: Orthogonality of spherical harmonics

Show that $\langle 1\,1|0\,0\rangle = 0$ (different m) and that $\langle 1\,0|0\,0\rangle = \langle 1\,0|2\,0\rangle = 0$ (same m, different l).

Solution. Write $Y_{lm} = \Theta_{lm}(\theta)\, e^{im\phi}$ with the normalisation

$$Y_{lm} = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}}\, P_l^m(\cos\theta)\, e^{im\phi}.$$

- *Different m.* The azimuthal integral $\int_0^{2\pi} e^{i(m'-m)\phi}\mathrm{d}\phi = 2\pi\delta_{mm'}$, so $\langle 1\,1|0\,0\rangle = 0$.
- *Same m, different l.* Let $x = \cos\theta$. Then $\langle 1\,0|0\,0\rangle \propto \int_{-1}^{1} x\,\mathrm{d}x = 0$ and $\langle 1\,0|2\,0\rangle \propto \int_{-1}^{1} x\,\tfrac{1}{2}(3x^2-1)\,\mathrm{d}x = 0$, because the integrands are odd.

For the $l = 2$ state with $m = l - 1 = 1$ one has explicitly

$$Y_{21} = -\sqrt{\frac{15}{8\pi}}\,\sin\theta\cos\theta\, e^{i\phi}.$$